

ASSIGNMENT UNIT 5

Conceptual Questions

- Q1.** A convolutional encoder has constraint length 3. A student says number of states = 3. Verify and correct. (BTL: AN)
- Q2.** Code rate is $1/2$, but output sequence length is not exactly double input. Explain why (consider termination). (BTL: AN)
- Q3.** Two encoders:
Encoder A: $K=3$, Encoder B: $K=5$.
Which has better error performance and why? (BTL: EV)
- Q4.** A received sequence perfectly matches a valid path in trellis. Can error still exist? (BTL: AN)
- Q5.** A system switches from hard decision to soft decision decoding. What numerical improvement is expected? (BTL: AN)

Analytical and Viterbi Problems

- Q6.** Convolutional encoder $(2,1,3)$, generator $(7,5)$, input 1011.
Calculate:
- Encode sequence
 - Draw partial trellis
 - Identify possible competing paths (BTL: AN)

Q7. Received sequence: 111000 (possibly corrupted), given encoder (7,5).

Calculate:

- Construct trellis
- Compute path metrics
- Find most likely input using Viterbi (BTL: AN)

Q8. A convolutional encoder produces output, but first few bits are missing.

Calculate:

- Reconstruct possible input sequences
- Justify ambiguity (BTL: CR)

Q9. Compare two decoding paths numerically: Path A metric = 3, Path B metric = 2.

Calculate:

- Which survives?
- What if soft decision used? (BTL: AN)

Q10. Encoder (3,1,3) with input 1101.

Calculate:

- Encode sequence
- Construct trellis
- Identify error correction capability (BTL: AN)